Phase Shifting and Reactance Compensation for Broadband Class E Amplifiers

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1 Introduction

Resonant amplifiers have an advantage over hard-switching converters by being able to have faster output response without sacrificing efficiency by acheiving zero volt switching (ZVS). However, resonant amplifiers inherently have more issues revolving around high circulating current, parasitics, load mismatch, potentially causing distorted waveforms and loss of ZVS. The class E amplifier is a theoretically 100% efficient, resonant amplifier that acheives ZVS using a shunt capacitance across a single switch in parallel with the resonant tank. However, a class E amplifier with high Q is extremely narrowband, whereas with low Q would be broadband but low efficiency from harmonic distortion. This paper discusses an efficient method to combine two, single switch class E amplifiers and provide reactance compensation to a highly variable load, in this case, plasma.

2 Theory